

torch.signal.windows.general_hamming#

https://pytorch.org/docs/stable/generated/torch.signal.windows.general_hamming.html

Description:

Computes the general Hamming window. The general Hamming window is defined as follows: The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window. alpha (float, optional) – the window coefficient. Default: 0.54.

Function Signature

```
torch.signal.windows.general_hamming(M, *, alpha=0.54,
sym=True, dtype=None, layout=torch.strided, device=None,
requires_grad=False)[source]#
```

Parameters

Keyword Arguments

Return type

Parameters

Keyword

alpha (float, optional) – the window coefficient. Default: 0.54. sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric Hamming window with the general
Hamming window.
>>> torch.signal.windows.general_hamming(10, sym=True)
tensor([0.0800, 0.1876, 0.4601, 0.7700, 0.9723, 0.9723, 0.7700,
0.4601, 0.1876, 0.0800])

>>> # Generates a periodic Hann window with the general Hamming
window.
>>> torch.signal.windows.general_hamming(10, alpha=0.5,
sym=False)
tensor([0.0000, 0.0955, 0.3455, 0.6545, 0.9045, 1.0000, 0.9045,
0.6545, 0.3455, 0.0955])
```

Detailed Documentation

Documentation

Computes the general Hamming window.

The general Hamming window is defined as follows:

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True.

M (int) – the length of the window. In other words, the number of points of the returned window.

`alpha` (float, optional) – the window coefficient. Default: 0.54.

`sym` (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`).

`layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

`device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: False.